



Report

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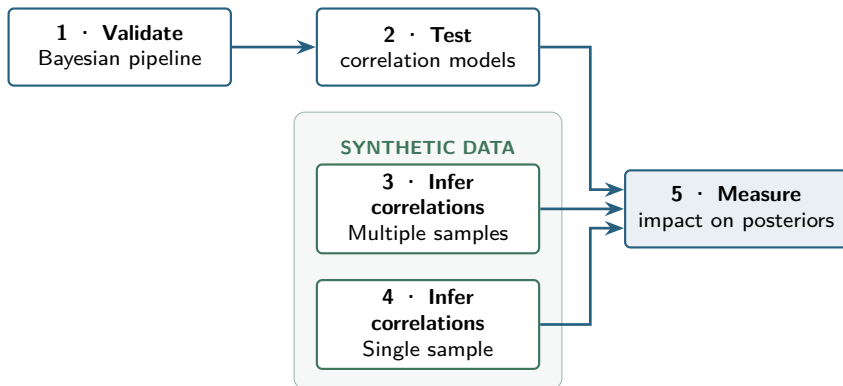
Summary

1 Tasks Status

Conference Materials Status

Task	Status	
2026-10 Conference	○	Simpósio IFUSP Presentation
2026-10 Conference	○	AGORÁ Presentation
2026-11 Poster	○	3rd School QC Poster
2026-12 Conference	○	LASNPA Presentation
2026-12 Paper	○	LASNPA Proceedings
2027-03 Conference	○	Quark Matter Presentation

Study Phases – Dependencies



Phase 1 – Bayesian Pipeline Validation

Task	Status	Output
↪ Reproduce Mankolli et al. results with the original JETSCAPE emulator	✓	Reference
↪ Reproduce the same results with the <code>hic_bayes</code> emulator	○	Validation
↪ Emulator closure test	○	Validation
↪ MCMC convergence test	○	Diagnostics
↪ Verify that model parameters are constrained	○	Posteriors

Phase 2 – Correlation Model Sensitivity

Task	Status	Output
↪ Au-Au-200 + d-Au-200: Diagonal vs. CS vs. RBF correlation models		Posteriors
↪ Compare posterior shifts, credible intervals, and parameter correlations		Analysis
↪ Question to answer: What is the appropriate correlation-length unit for an RBF covariance spanning heterogeneous observables/bins?		Study
↪ Quantify and compare the emulator-error and experimental-error covariance matrices in the likelihood		Analysis

Phase 3 – Correlation Inference from Multiple Samples

Task	Status	Output
↪ Generate controlled synthetic datasets with known covariance (N samples; goal: $N = 10$)		Dataset
↪ Classical correlation inference		Benchmark
↪ Quantum-learning correlation inference		Benchmark
↪ Classical vs. quantum comparison against the known covariance		Comparison

Phase 4 – Single-Sample Correlation Inference

Task	Status	Output
↪ Study correlation approximations when only one experimental sample is available		Heuristics
↪ Use reported uncertainty components to constrain plausible correlation structures		Models

Phase 5 – End-to-End Bayesian Impact

Task	Status	Output
↪ Propagate inferred covariance matrices through the Bayesian likelihood • Single sample: CS, RBF, Zero, GP • Multiple samples: Classical, Quantum	○	Posteriors
↪ Quantify covariance-estimation error → posterior error	○	Final study